

William Lim

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EDUCATION

California State University, Fullerton

May 2026

B.S. in Computer Science, B.A. in Studio Art

Relevant Coursework: Compilers & Languages, Operating Systems, Assembly, Computer System Architecture, Python, Scientific Writing, Algorithm Engineering, Back-End Engineering, Cybersecurity, File Structures & Databases, Multivariable Calculus, Statistics, Physics, Artificial Intelligence, Machine Learning, Edge AI, Software Architecture, Modern Software Deployment and Operations, Linear Algebra, Discrete Math, Design, Illustration

TECHNICAL SKILLS

Languages: Python, C, C++, Java, C#, Intel x86 and RISC-V assembly, TypeScript, HTML/CSS, SQL, R, Prolog, PHP, Rust

Frameworks, Skills & Tools: AWS (Serverless/Autoscaling, Caching, security, etc.), Terraform (and Packer), React, Next, Vue, Node.js, Prisma, SQLAlchemy, vector databases, FastAPI, Flask, Docker, GitHub Actions, PyTorch, (Geo)Pandas, Matplotlib, NumPy, Scikit-Learn, OpenCV, Linux, OpenGL, WebGL2, Media Pipe, ANTLR4, Ollama, Gemini API, SEO, MCP, Coverage and Unit Testing, SSH, Firewalls, Least Privilege, AES, RSA, Networking, Wire Shark, Agile, Scrum, Kanban, MongoDB, DynamoDB

EXPERIENCE

Web Designer, *Ride With Care Ambulette Transportation*

Aug. 2024 – Jan. 2025

Web Designer, *ExecPathfinders LLC*

July 2023 – Oct. 2024

Software Engineer Intern, *Garrett Integrated Automotive Corp. (GIAC)*

Jan. 2024 – Jan. 2024

Developed proprietary software for querying ISAM databases, heightening data retrieval efficiency.

June 2022 – Aug. 2022

Developed REST API for product data queries. Developed internal data portal for assisting customer support and a reactive embed for business partners' websites utilizing the REST API.

PROJECTS | *Code can be found on GitHub.*

RISC-V CPU Emulator and Assembler

Nov. 2025

Emulates a thorough subset of the RV32I ISA and RV32F extension using bit-level connections and gate-level implementations. Includes proprietary RISC-V assembler.

A-Star GPS Road Navigation System | *Navigates Open Street Map Data Using A**

Dec. 2025

Using A* and UCS on OpenStreetMap data, this GPS software navigates optimally to various geological coordinates in 65% of tests. The system accounts for traffic laws, road junctions, signage, and lane counts.

Vanta 3D Engine | *A highly portable 3D render engine with native WebGL2 support*

Jan. 2026 – Present

Includes a WebGL2 Node/Scene backend with a Physically Based Rendering pipeline. A native Vulkan backend is also being developed.

Python to Intel X86 Assembly Compiler and Bindings

Dec. 2024 – Sep. 2025

Compiles and assembles Python source code into statically typed x86 Assembly via the Python AST and proprietary x86 Assembly bindings.

OpenCV/AI Demo | *Emotion Recognition, Face Tracking, and Eye Tracking Web Demo*

Jan. 2025 – Feb. 2025

A demo showcasing my own convolutional neural network for face-image emotion recognition, face landmark tracking via Media Pipe, and eye tracking using image processing techniques with OpenCV.

LEADERSHIP EXPERIENCE

Board Member, Association for Computing Machinery (ACM) Algorithms

Aug. 2025 – Present

Board Member, Association for Computing Machinery Game Development

Aug. 2025 – Jan. 2026

California State University, Fullerton

- Prepared and presented lectures to Computer Science students on basic to advanced CS concepts.
- Organized, led, and performed successful outreach attracting new members to ACM, the largest CSUF student organization with over 2500 members.

Participant/Group Leader, UCLA Glitch Club Build with Gemini Google Hackathon

Mar. 2026